

Edge Case Analysis Report

Model accuracy on 77 hand-curated edge-case pairs, sub-divided by their integer ground truth label. Edge cases are pairs where the correct classification is non-obvious, requires deep game knowledge, or involves conditional/context-dependent interactions.

Accuracy = fraction of pairs where model predicted the correct integer GT label.
F1 (target) = F1 for the expected class specifically.
Macro F1 = unweighted average across all classes present in the sub-mask.

Edge Positive (n=30) — edge cases where GT = +1

These are pairs the mask flagged as edge cases that *are* genuine synergies. Models must correctly identify the positive interaction despite the non-obvious or conditional nature of the combo.

Pair List

Card A	Card B	Int GT
Armaments	Blood for Blood	+1
Armaments	Reaper	+1
Bash	Havoc	+1
Battle Trance	Second Wind	+1
Berserk	Body Slam	+1
Berserk	True Grit	+1
Brutality	Second Wind	+1
Burning Pact	Second Wind	+1
Burning Pact	Sentinel	+1
Corruption	Power Through	+1
Corruption	Second Wind	+1
Dark Embrace	Warcry	+1
Dual Wield	Second Wind	+1
Evolve	Havoc	+1
Evolve	Second Wind	+1
Fire Breathing	Dark Embrace	+1
Fire Breathing	Evolve	+1
Headbutt	Fire Breathing	+1
Offering	Corruption	+1
Pommel Strike	Second Wind	+1
Power Through	Brutality	+1
Rampage	Havoc	+1
Seeing Red	True Grit	+1
Sentinel	Corruption	+1
Shockwave	Blood for Blood	+1

Shockwave	Reaper	+1
Shrug it Off	Second Wind	+1
True Grit	Sentinel	+1
True Grit	Sever Soul	+1
Warcry	Fire Breathing	+1

Model Performance

Model	n	Accuracy	Prec	Recall	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	30	60.0%	100.0%	60.0%	75.0%	37.5%	0 (0%)	12 (40%)	18 (60%)
gpt54	30	53.3%	100.0%	53.3%	69.6%	23.2%	3 (10%)	11 (37%)	16 (53%)
gemini10pro	30	66.7%	100.0%	66.7%	80.0%	20.0%	2 (7%)	6 (20%)	20 (67%)
gemini15flash	30	33.3%	100.0%	33.3%	50.0%	16.7%	1 (3%)	19 (63%)	10 (33%)
gpt4omini	30	50.0%	100.0%	50.0%	66.7%	22.2%	3 (10%)	12 (40%)	15 (50%)
gpt4ominit	30	36.7%	100.0%	36.7%	53.7%	17.9%	2 (7%)	17 (57%)	11 (37%)
gemini3flash	30	76.7%	100.0%	76.7%	86.8%	28.9%	1 (3%)	6 (20%)	23 (77%)
llama4maverick	30	73.3%	100.0%	73.3%	84.6%	28.2%	1 (3%)	7 (23%)	22 (73%)

Edge Neutral (n=43) — edge cases where GT = 0

Pairs flagged as edge cases that are ultimately neutral.
The risk here is false positives — models that over-pattern-match on card interactions may incorrectly predict synergy where there is none.

Pair List

Card A	Card B	Int GT
Anger	Dropkick	+0
Barricade	Berserk	+0
Barricade	Corruption	+0
Barricade	Sever Soul	+0
Bash	Double Tap	+0
Berserk	Barricade	+0
Bludgeon	Feed	+0
Brutality	Power Through	+0
Carnage	Exhume	+0
Corruption	Feed	+0
Corruption	Fiend Fire	+0
Corruption	Flex	+0
Corruption	Offering	+0
Corruption	Rage	+0
Demon Form	Double Tap	+0
Double Tap	Fiend Fire	+0
Evolve	Sever Soul	+0
Exhume	Exhume	+0

Fiend Fire	Pommel Strike	+0
Fiend Fire	Shrug it Off	+0
Fire Breathing	Fiend Fire	+0
Fire Breathing	Sever Soul	+0
Ghostly Armor	Exhume	+0
Immolate	Havoc	+0
Inflame	Double Tap	+0
Juggernaut	Berserk	+0
Juggernaut	Corruption	+0
Juggernaut	Sever Soul	+0
Limit Break	Double Tap	+0
Offering	Clash	+0
Rage	Double Tap	+0
Rampage	Warcry	+0
Reckless Charge	Havoc	+0
Reckless Charge	True Grit	+0
Seeing Red	Warcry	+0
Sentinel	Dual Wield	+0
Sentinel	Feel No Pain	+0
Sentinel	Warcry	+0
Shockwave	Double Tap	+0
Spot Weakness	Double Tap	+0
Uppercut	Double Tap	+0
Warcry	Dropkick	+0
Wild Strike	Havoc	+0

Model Performance

Model	n	Accuracy	Prec	Recall	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	43	58.1%	100.0%	58.1%	73.5%	24.5%	4 (9%)	25 (58%)	14 (33%)
gpt54	43	62.8%	100.0%	62.8%	77.1%	25.7%	8 (19%)	27 (63%)	8 (19%)
gemini10pro	43	20.9%	100.0%	20.9%	34.6%	11.5%	6 (14%)	9 (21%)	28 (65%)
gemini15flash	43	62.8%	100.0%	62.8%	77.1%	25.7%	3 (7%)	27 (63%)	13 (30%)
gpt4omini	43	44.2%	100.0%	44.2%	61.3%	20.4%	5 (12%)	19 (44%)	19 (44%)
gpt4ominift	43	72.1%	100.0%	72.1%	83.8%	27.9%	5 (12%)	31 (72%)	7 (16%)
gemini3flash	43	48.8%	100.0%	48.8%	65.6%	21.9%	13 (30%)	21 (49%)	9 (21%)
llama4maverick	43	48.8%	100.0%	48.8%	65.6%	21.9%	2 (5%)	21 (49%)	20 (47%)

Edge Negative (n=4) — edge cases where GT = -1

Pairs flagged as edge cases that are anti-synergies.
Small sample (n=4). Treat results as illustrative only.

Pair List

Card A	Card B	Int GT
Battle Trance	Warcry	-1
Offering	Sever Soul	-1
Offering	True Grit	-1
True Grit	Armaments	-1

Model Performance

Model	n	Accuracy	Prec	Recall	F1 (target)	Macro F1	Pred -1	Pred 0	Pred +1
gpt4o	4	75.0%	100.0%	75.0%	85.7%	42.9%	3 (75%)	1 (25%)	0 (0%)
gpt54	4	50.0%	100.0%	50.0%	66.7%	22.2%	2 (50%)	1 (25%)	1 (25%)
gemini10pro	4	25.0%	100.0%	25.0%	40.0%	20.0%	1 (25%)	0 (0%)	3 (75%)
gemini15flash	4	50.0%	100.0%	50.0%	66.7%	33.3%	2 (50%)	2 (50%)	0 (0%)
gpt4omini	4	25.0%	100.0%	25.0%	40.0%	20.0%	1 (25%)	0 (0%)	3 (75%)
gpt4ominift	4	25.0%	100.0%	25.0%	40.0%	13.3%	1 (25%)	2 (50%)	1 (25%)
gemini3flash	4	50.0%	100.0%	50.0%	66.7%	22.2%	2 (50%)	1 (25%)	1 (25%)
llama4maverick	4	75.0%	100.0%	75.0%	85.7%	42.9%	3 (75%)	0 (0%)	1 (25%)

Warning — small sample (n=4). Aggregate metrics are unreliable; the pair list above is more informative.

Summary

Model	Pos acc (n=30)	Neu acc (n=43)	Neg acc (n=4)
gpt4o	60.0%	58.1%	75.0%
gpt54	53.3%	62.8%	50.0%
gemini10pro	66.7%	20.9%	25.0%
gemini15flash	33.3%	62.8%	50.0%
gpt4omini	50.0%	44.2%	25.0%
gpt4ominift	36.7%	72.1%	25.0%
gemini3flash	76.7%	48.8%	50.0%
llama4maverick	73.3%	48.8%	75.0%

Generated by `edge_case_analysis.py`